The 043 Training Setup

A design review: one model over expression, protein, isolate, segregant and promoter readouts, its objective, and how a transfer gain is decided

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October 23, 2026

Abstract

Experiment 043 trains one model over nine datasets of the 043 store: deletion transcriptomes and proteomes, natural-isolate transcriptomes and proteomes, segregant single-cell expression, and a promoter library read by sort-seq. Every gene enters as a composite of four frozen sequence embeddings plus a position term, a trunk encodes each distinct reference context once, a perturbation operator applies the record's marks by cross-attention, and two heads read a gene-level and a record-level value, with the dataset entering only at the heads. The objective is a binned regression (HL-Gauss) on per-dataset standardized targets, averaged within each dataset and weighted across datasets. Written in the notation of the Nature Biotechnology manuscript, the setup is given equation by equation, followed by the evaluation protocol and the rule that decides whether joint training transfers to a dataset, the single-seed overnight runs of 2026-10-22 against the linear bars, and the design questions left open. Under the uniform record sampler the promoter library holds 84.1% of the training records and carries an expected 0.573 of the batch objective's weight; the step-matched transfer on Kemmeren 2014 read −0.032, +0.009 and −0.020 for the three complete joint runs, one seed each and below the decision rule.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 Notation ✗

The symbols are the manuscript's (Methods, Table "Notation", and Eqs. for the encoder, the perturbation operator, the readouts and the loss), reused with the same meaning. Where 043 needs more, the symbol is defined once in the second block of Table 1 and marked "043". Four manuscript symbols carry a narrower meaning here and say so in their rows: the task index becomes the dataset index s, the perturbation tokens are the record's marks, the type τ_t is a change type, and the mask m marks measured entries only, since 043 marks carry no magnitude. Two manuscript symbols whose objects 043 does not build are reused: s, the gene-set index of the morphology head, and λ, the graph-prior weights, absent because 043 masks attention instead of regularizing it. Within each block the rows are alphabetical by the symbol's letter name.

Table 1: Notation. First block: manuscript symbols as 043 uses them. Second block: symbols 043 adds. Shapes are per record unless a batch dimension is written. Numbers are the values of the overnight runs of 2026-10-22 (`conf/train_043_default.yaml` with `data.batch_size=8`).

Symbol	Meaning in 043	Source
$a,\ h$	attention head index, number of heads; $h = 9$ in the trunk and the operator	paper
$A^{(k)}$	relation-k adjacency over genes; nine relations, one per head of the masked trunk layer, used as a hard mask (Eq. 6), not as a KL prior	paper
$\alpha^{(\ell,a)}$	trunk self-attention map of layer ℓ, head a	paper
B	records per batch; 8 in every overnight run (the config default is 16)	paper
$\beta^{a}_{i,t}$	operator cross-attention weight of query i on mark t in head a; the softmax also runs over the null key $\varnothing$	paper
$d,\ d_k$	model width $d = 144$; head width $d_k = d/h = 16$	paper
Embed, $\mathbf{E}$	entity encoder; learned lookup table. Every learned lookup of 043 is an $\mathbf{E}$ with a superscript naming it: $\mathbf{E}^{\text{gap}}$, $\mathbf{E}^{\text{chr}}$, $\mathbf{E}^{\text{pgap}}$, $\mathbf{E}^{\text{par}}$, $\mathbf{E}^{\text{type}}$, $\mathbf{E}^{\text{data}}$, $\mathbf{E}^{\text{env}}$	paper
ε	environment; enters only through the token $\mathbf{c}^{\text{env}}$ at the heads	paper
F_θ	trunk (encoder), $H_c = F_\theta(G_c)$	paper
FFN, LN, MLP	position-wise feed-forward network, layer normalization, two-layer perceptron (Linear, GELU, dropout 0.1, Linear)	paper
$G,\ g_i,\ N$	reference cell; gene i of the shared index; $N = 6{,}607$	paper
$H,\ h_i,\ h_{\text{CLS}}$	trunk states $(h_{\text{CLS}}, h_1, \ldots, h_N) \in \mathbb{R}^{(N+1)\times d}$; $H^{(\ell)}$ after layer ℓ	paper
$H_{\text{pert}},\ h_i^{\text{pert}}$	operator output, same shape as H	paper
$\ell,\ L$	trunk layer index, number of layers; $L = 4$	paper
$\mathcal{L},\ \mathcal{L}_y$	loss; supervised objective of a batch (Eq. 21). 043 has no graph term, so $\mathcal{L} = \mathcal{L}_y$	paper
$m_{b,i}$	1 when entry i of record b is measured, else 0 (the manuscript's $m_t^{(b)}$)	paper
M_b	number of marks of record b (the manuscript's M)	paper
$p,\ \mathbf{p}_t,\ \mathbf{P}$	perturbation set; its tokens are the record's marks $t = 1, \ldots, M_b$ (Eq. 7)	paper
$\mathcal{R}_\phi$	decoder: the gene head and the record head (Eqs. 13, 14)	paper
$\mathcal{T}_\psi$	perturbation operator (Eqs. 8, 9)	paper
$\tau_t,\ \mathbf{t}_\tau$	type of mark t, here a change type $\tau_t \in \mathcal{C}$; its learned embedding	paper
Θ	all trainable parameters, 1.85 M in the overnight runs (job 6136 log)	paper
W_Q, W_K, W_V, W_O	query, key, value and output maps	paper
w_s	weight of dataset s in the objective (the manuscript's task weight w_t)	paper
$x_i,\ x_{\text{CLS}}$	input token of gene i (Eq. 4); learned CLS token	paper
$y,\ \hat{y}$	released label, prediction	paper
$\bar{\mathbf{k}}_b$	mean of the mark keys of record b (the primer's z_S, renamed because z is the target here); zero without marks	043
$\mathbf{b}_s^{\text{gene}},\ \mathbf{b}_s^{\text{rec}}$	learned per-dataset bias of the gene and the record head, $\in \mathbb{R}^{n_{\text{out}}}$	043
$\mathcal{B},\ n_s(\mathcal{B}),\ \mathcal{S}_\mathcal{B}$	a batch of records; the records of dataset s in it; the datasets with a measured entry in it	043
$c,\ G_c,\ H_c$	reference context (S288C, or an isolate under route A); its cell; its trunk states	043
$\mathbf{c}_b$	conditioning token $[\mathbf{c}^{\text{type}}; \mathbf{c}^{\text{data}}; \mathbf{c}^{\text{env}}] \in \mathbb{R}^{40}$ of record b (Eq. 12)	043
$\mathcal{C}$	change-type vocabulary: reference, deletion, variant, loss of function, absent, present, promoter	043
χ	node of the experiment-type path (Eq. 11)	043
Δ	HL-Gauss bin width, $8/32 = 0.25$ in z	043
$\eta_{s,i},\ \mu_s,\ \sigma_s$	per-gene center (zero unless the dataset is centered per gene), pooled mean and standard deviation of dataset s over its training rows	043
Four(ξ)	Fourier features $[\sin(2\pi 2^f \xi), \cos(2\pi 2^f \xi)]_{f=0}^{7} \in \mathbb{R}^{16}$	043
γ_u	quantile level $u/20$, $u = 1, \ldots, 19$	043
$\kappa,\ \mathcal{K}$	typed-gap kind of a block slot; the 12 kinds of `GapKind` (no allele, locus gap, short flank, …, ambiguous sequence)	043
$\mathbf{k}_\varnothing$	learned null key of the operator, one per head, with value zero	043
$\lambda_0, \lambda_1, \lambda_2$	coefficients of the fitted doubles rung (Eq. 27)	043
$n_{\text{bin}},\ n_q,\ n_{\text{out}}$	HL-Gauss bins (32); pinball quantiles (19); head output width (n_{bin}, n_q or 1)	043
$n_s,\ n$	training records of dataset s; $n = \sum_s n_s = 50{,}767$	043
$n_{\text{ep}},\ n_{\text{step}}$	epochs and optimizer steps of a run	043
$\nu_{c,i}^{\text{chr}},\ \nu_{c,i}^{\text{gene}}$	copy number of gene i's chromosome relative to the basal ploidy, and of the gene, in context c	043
o_t	allele parent of mark t (segregants; row 0 is "no parent")	043
$\hat{\mathbf{o}}_{b,i}$	head output for entry i of record b, $\in \mathbb{R}^{n_{\text{out}}}$	043

Symbol	Meaning in 043	Source
$\omega_u(z),\ \hat{\omega}_u$	target and predicted mass of HL-Gauss bin u	043
$q,\ \mathcal{Q}$	frozen block index, $\mathcal{Q} = \{\text{up}, \text{down}, \text{prot}, \text{calm}\}$	043
$s,\ \mathcal{S},\ s(b)$	dataset index; the training datasets of an arm ($\lvert\mathcal{S}\rvert = 9$ for `joint_9`); the dataset of record b	043
stat_e	an evaluation statistic at epoch e; $\overline{\text{stat}}$ its registered-window mean	043
tr_s	pre-standardization transform of dataset s	043
u	output component index (bin or quantile)	043
$v_u,\ \bar{v}_u$	HL-Gauss bin edges and centers	043
$\mathcal{W}$	registered window: the epochs whose evaluations are averaged	043
$x_{c,i}^{(q)},\ d_q$	frozen vector of block q for gene i in context c; its width, $(768, 768, 1024, 768)$	043
ξ_i	relative midpoint of gene i on its chromosome, in $[0, 1]$	043
$z_{b,i},\ \hat{z}_{b,i}$	standardized target (Eq. 15); point prediction in z	043
$\zeta_{\text{attn}},\ \zeta_{\text{ffn}}$	ReZero gates of the operator, initialized at zero	043

2 Data and model flow ✗

Figure 1 follows a record from the raw store to the number the decision rule reads. The store and its datasets are described in the 043 datasets primer (`notes-tex/multimodal/043-datasets-primer/`, Secs. 1 to 4 and 6); what follows starts where the primer stops, at the representation the trainer consumes. Two facts set the shape of the pipeline. First, the representation holds keys, not vectors: each gene slot of a reference context carries one key per frozen block, or a typed gap, and the trainer reads the vectors of exactly the keys its records use, failing on a missing one. Second, the reference context is part of the input: a deletion, segregant or promoter record is read on S288C, an isolate on its own alleles (route A), so a batch carries between one and B distinct contexts, and the trunk runs once per distinct context.

Stages. The global split assigns every record one role before anything trains (Sec. 5.1). The prepare stage (`train_043.py --stage prepare`, job 6134) reads every record of the arm's datasets with its route-A representation, drops the test records, applies the per-dataset transform tr_s and writes keys, marks, contexts, targets and the nine gene–gene edge lists into one store. Training fits the standardization on the training rows, builds the model, and logs the training objective every 50 steps and the validation pass every evaluation epoch. Nothing downstream of the prepared store reads fold 0.

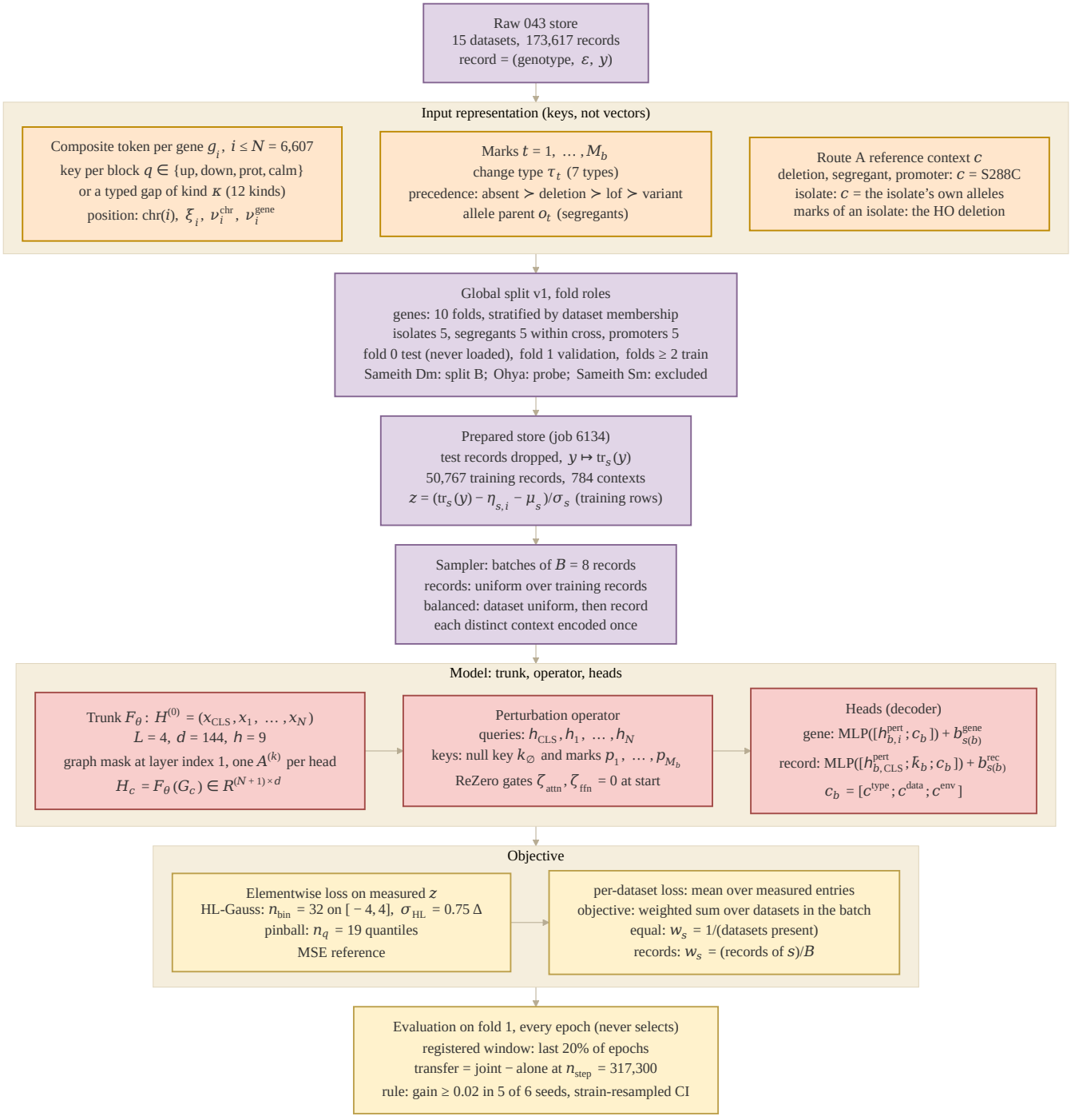


Figure 1: The 043 training pipeline, from the raw store to the registered-window read. Purple, data: the store, the fold roles of the global split, the prepared store (job 6134; test records dropped before any label is loaded) and the batch. Orange, input side: the composite token per gene with its position features, the marks with their change types and allele parents, and the route-A reference context. Red, model: trunk, perturbation operator and heads. Yellow, objective and evaluation. The 784 contexts of the prepared store are S288C and the 783 isolates outside fold 0. In the rendered math, vectors written bold in the text ($\mathbf{c}_b$, $\mathbf{p}_t$, $\bar{\mathbf{k}}_b$) appear in the regular face. Rendered 212 mm wide and placed at 180 mm, a scale of 0.85.

3 Model, step by step ✗

The model keeps the manuscript's factorization $\hat{f}_\theta = \mathcal{R}_\phi \circ \mathcal{T}_\psi \circ F_\theta$ (Methods, "Problem setup") and changes what enters it: the input token of a gene is a function of the reference context, the perturbation tokens are typed marks, and the decoder is two shared heads with the dataset entering only at their input and output. Shapes are given for one record unless a batch axis is written; C is the number of distinct contexts in a batch.

3.1 The gene token ✗

Step 1, the four frozen blocks. For context c and gene i, each block $q \in \mathcal{Q}$ holds either a key or a typed gap $\kappa \in \mathcal{K}$. A key indexes a frozen fp16 vector $x_{c,i}^{(q)} \in \mathbb{R}^{d_q}$: the FUDT upstream encoder on the 1,003 bp window ($d_{\text{up}} = 768$), the FUDT downstream encoder on the 300 bp window ($d_{\text{down}} = 768$), ProtT5-XL on the protein ($d_{\text{prot}} = 1024$) and CaLM on the coding sequence ($d_{\text{calm}} = 768$). Each block is normalized and projected separately,

$$\tilde{x}_{c,i}^{(q)} = \begin{cases} W^{(q)}\,\mathrm{LN}^{(q)}\big(x_{c,i}^{(q)}\big) + b^{(q)} & \text{if block } q \text{ of gene } i \text{ has a key in } c, \\ \mathbf{E}^{\text{gap}}_{q,\kappa} & \text{if it carries the gap } \kappa, \end{cases} \qquad W^{(q)} \in \mathbb{R}^{d \times d_q}, \tag{1}$$

so a gap is a learned vector of its own kind and block and is never filled with another sequence's vector. The vectors stay frozen; $\mathrm{LN}^{(q)}$, $W^{(q)}$, $b^{(q)}$ and $\mathbf{E}^{\text{gap}} \in \mathbb{R}^{(4\cdot 12)\times d}$ train. The overnight runs held 929,348, 326,255, 329,430 and 514,189 rows of the four tables on the card, 3.39 GB (job 6136 log).

Step 2, the composite. The four projections are summed,

$$\tilde{x}_{c,i} = \sum_{q \in \mathcal{Q}} \tilde{x}_{c,i}^{(q)} \in \mathbb{R}^d. \tag{2}$$

Step 3, the position term. Gene i lies on chromosome $\mathrm{chr}(i) \in \{0, \dots, 16\}$ (0 is the mitochondrion) at relative midpoint $\xi_i \in [0, 1]$ of the reference. With $\nu_{c,i}^{\text{chr}}$ the copy number of that chromosome relative to the basal ploidy (1 except on aneuploid isolate chromosomes) and $\nu_{c,i}^{\text{gene}}$ the gene's copy number,

$$\mathrm{pos}_{c,i} = \nu_{c,i}^{\text{chr}}\,\mathbf{E}^{\text{chr}}_{\mathrm{chr}(i)} + W_\xi\,\mathrm{Four}(\xi_i) + b_\xi + \mathbf{w}_\nu\big(\log_2(\nu_{c,i}^{\text{gene}} + 1) - 1\big) + \mathbf{b}_\nu + \mathbf{E}^{\text{pgap}}_{\kappa'_{c,i}}, \tag{3}$$

with $\mathbf{E}^{\text{chr}} \in \mathbb{R}^{19\times d}$ (17 chromosomes and two rows for off-graph tokens, an accessory ORF and the reporter construct), $W_\xi \in \mathbb{R}^{d\times 16}$ and $\mathbf{w}_\nu, \mathbf{b}_\nu \in \mathbb{R}^d$. The copy feature is 0 at one copy and -1 at zero copies. An unknown copy number (a ploidy the source did not report, a value the pangenome matrix does not release) drops its term and selects the learned position-gap row $\kappa'_{c,i}$ of $\mathbf{E}^{\text{pgap}} \in \mathbb{R}^{4\times d}$ (row 0 when nothing is unknown).

Step 4, the input token. The manuscript's node feature $x_i = \mathrm{Embed}(g_i)$ becomes context dependent,

$$x_{c,i} = \tilde{x}_{c,i} + \mathrm{pos}_{c,i}, \qquad H_c^{(0)} = (x_{\text{CLS}}, x_{c,1}, \dots, x_{c,N}) \in \mathbb{R}^{(N+1)\times d}. \tag{4}$$

For S288C every slot has its reference keys; under route A an isolate's slots carry the isolate's own alleles, with the genes outside the released allele set and the genes the record calls absent as gaps.

3.2 The trunk ✗

Step 5, the layers. F_θ is four layers of width $d = 144$ with $h = 9$ heads of width $d_k = 16$. The layer is the manuscript's Eqs. for the queries, keys, values and attention map, but post-normalized, where the manuscript writes the pre-normalized form,

$$\hat{H}^{(\ell)} = \mathrm{LN}\Big(H^{(\ell-1)} + \mathrm{Drop}\big(\big[O^{(\ell,1)} \| \cdots \| O^{(\ell,h)}\big] W_O^{(\ell)}\big)\Big), \qquad H^{(\ell)} = \mathrm{LN}\Big(\hat{H}^{(\ell)} + \mathrm{Drop}\big(\mathrm{FFN}^{(\ell)}(\hat{H}^{(\ell)})\big)\Big). \tag{5}$$

Step 6, the graph mask. In layer $\ell = 2$ (`mask_layers: [1]`, counted from zero) head a may attend only along graph $k(a)$; the other three layers are unmasked. With $A^{(k)}$ symmetrized,

$$\alpha^{(2,a)} = \mathrm{softmax}\Big(\frac{Q^{(2,a)}(K^{(2,a)})^\top}{\sqrt{d_k}} + \Gamma^{(a)}\Big), \qquad \Gamma_{ij}^{(a)} = \begin{cases} 0 & \text{if } A_{ij}^{(k(a))} = 1,\ A_{ji}^{(k(a))} = 1,\ i = j, \text{ or } i \text{ or } j \text{ is the CLS}, \\ -\infty & \text{otherwise}, \end{cases} \tag{6}$$

with $k(0), \ldots, k(8)$ the physical, regulatory and TFLink graphs and the STRING 12.0 neighborhood, fusion, co-occurrence, coexpression, experimental and database channels. The objective carries no Ω_k term: 043 takes the manuscript's $\lambda \to \infty$ limit of graph-regularized attention at one layer and leaves the rest free.

Step 7, one encode per context. Each distinct context of a batch is encoded once, $H_c = F_\theta(G_c) \in \mathbb{R}^{(N+1)\times d}$, a $[C, N+1, d]$ tensor, and record b reads $H_{c(b)}$. The trunk trains, so nothing is cached across batches.

3.3 Marks and the perturbation operator ✗

Step 8, change types. Each gene of a genotype has one change type in $\mathcal{C}$. When the source gives a gene several, the first in the order absent, deletion, loss of function, variant, present, promoter wins: a gene that is gone outranks a call on its sequence, which outranks a variant that is still expressed. A mark is a token the operator attends to: a deleted gene, a gene a segregant inherits from a non-BY parent (a variant, with that parent as o_t), the HO deletion of an isolate under route A, an isolate's present accessory ORF (off-graph), and Vaishnav's reporter, an off-graph token whose upstream key is the variant's 1,003 bp construct window. Under route A the isolate's variants live in its context, not in its marks. The overnight runs kept at most 1,024 marks per record (`data.mark_cap`: absent, deletion and loss-of-function marks first, then a fixed per-record random subset), a smoke-time truncation the design does not have; N'Guessan 2025 records carry a median of 2,910 marks (job 6120).

Step 9, the mark key. Mark t of record b has the composite and position of its gene in the record's context (or of its off-graph token, with the replicon row of $\mathbf{E}^{\text{chr}}$ and the Fourier term zeroed), plus its change type and allele parent,

$$\tilde{\mathbf{p}}_t = \tilde{x}_t + \text{pos}_t + \mathbf{t}_{\tau_t} + \mathbf{E}^{\text{par}}_{o_t}, \qquad \mathbf{p}_t = W_p\,\text{LN}(\tilde{\mathbf{p}}_t) + b_p + \mathbf{1}[\text{mark } t \text{ is on the graph}]\; h_{c(b),\,j_t}, \tag{7}$$

where j_t is the marked gene and $\mathbf{E}^{\text{par}}_0 = 0$ is "no parent". This is the manuscript's token $\mathbf{p}_t = r(e_t) + \mathbf{t}_{\tau_t} + m_t\mathbf{w}$ with two changes: the projected mark token is added for in-cell genes as well, and there is no magnitude term. $\mathbf{P}_b \in \mathbb{R}^{M_b\times d}$, padded to $[B, \max_b M_b, d]$.

Step 10, cross-attention with a null sink. Every token of the context, the CLS included, is a query. Each head adds a learned null key $\mathbf{k}_\varnothing$ with value zero, so a record without marks attends only to it and a record with one mark does not see a degenerate softmax,

$$\beta^a_{i,t} = \underset{t\in\{\varnothing,1,\ldots,M_b\}}{\text{softmax}} \frac{(h_i W^a_Q)\cdot \mathbf{k}^a_t}{\sqrt{d_k}}, \quad \mathbf{k}^a_t = \mathbf{p}_t W^a_K, \quad \Delta h_i = \Big[\textstyle\sum_{t\ge1}\beta^1_{i,t}\,\mathbf{p}_t W^1_V \,\Big\|\cdots\Big\|\, \sum_{t\ge1}\beta^h_{i,t}\,\mathbf{p}_t W^h_V\Big] W_O, \tag{8}$$

for $i \in \{\text{CLS}, 1, \ldots, N\}$; the null term contributes weight but no value.

Step 11, the ReZero residual. Where the manuscript's operator normalizes, 043 gates,

$$h'_i = h_i + \zeta_{\text{attn}}\,\text{Drop}(\Delta h_i), \qquad h^{\text{pert}}_i = h'_i + \zeta_{\text{ffn}}\,\text{Drop}\big(\text{FFN}(h'_i)\big), \qquad H_{\text{pert}} = \mathcal{T}_\psi(H_{c(b)}, p_b) \in \mathbb{R}^{(N+1)\times d}, \tag{9}$$

with $\zeta_{\text{attn}} = \zeta_{\text{ffn}} = 0$ at initialization, so at step 0 the operator is the identity and $H_{\text{pert}} = H_{c(b)}$. The record-level summary of the marks is

$$\bar{\mathbf{k}}_b = \frac{1}{\max(M_b, 1)} \sum_{t=1}^{M_b} \mathbf{p}_t \in \mathbb{R}^d. \tag{10}$$

3.4 Conditioning tokens and heads ✗

Step 12, the experiment-type token. Each dataset reads one phenotype class of the schema, and the class enters as the sum of one learned row per node of its path,

$$\mathbf{c}^{\text{type}}_b = \sum_{\chi\in\text{path}(\text{class}(s(b)))} \mathbf{E}^{\text{type}}_\chi \in \mathbb{R}^{16}, \tag{11}$$

where a path runs root, group, leaf: phenotype, expression, then the microarray (Kemmeren), RNA-seq (Caudal) or single-cell (Boocock, N'Guessan) class; phenotype, protein, then protein abundance (Messner, Zelezniak, Muenzner) or turnover (Muenzner turnover); phenotype, promoter, pooled sort-seq (Vaishnav). The path is the minimal differential subset of the ontology DAG for these six leaves: it keeps the one level that separates them into the three readout groups the heads share, so two leaves under one group share that group's row and differ by their own. The group nodes are modeling-side groups, not schema classes (`train_043.py`, `EXP_TYPE_PATHS`); the root is on every path and acts as a shared offset. The flat alternative, one row per leaf, is `exp_type_mode: flat` (Sec. 7).

Step 13, the conditioning token. The dataset and the environment are flat lookups,

$$\mathbf{c}_b = \left[\mathbf{c}^{\text{type}}_b;\ \mathbf{E}^{\text{data}}_{s(b)};\ \mathbf{E}^{\text{env}}_{\varepsilon(b)}\right] \in \mathbb{R}^{16+8+16=40}, \tag{12}$$

over the nine datasets and nine environments of the prepared store.

Step 14, the gene head. Every gene-level dataset shares one head that reads the gene's own perturbed state, the manuscript's equivariant readout with the conditioning token appended,

$$\hat{\mathbf{o}}_{b,i} = \text{MLP}_{\text{gene}}\left([h^{\text{pert}}_{b,i};\ \mathbf{c}_b]\right) + \mathbf{b}^{\text{gene}}_{s(b)} \in \mathbb{R}^{n_{\text{out}}}, \qquad i = 1, \ldots, N, \tag{13}$$

with $\text{MLP}_{\text{gene}} : \mathbb{R}^{184} \to \mathbb{R}^{128} \to \mathbb{R}^{n_{\text{out}}}$, a $[B_{\text{gene}}, N, n_{\text{out}}]$ output.

Step 15, the record head. Vaishnav's one value per promoter is read from the perturbed CLS, the mark summary and the conditioning token,

$$\hat{\mathbf{o}}_b = \text{MLP}_{\text{rec}}\left([h^{\text{pert}}_{b,\text{CLS}};\ \bar{\mathbf{k}}_b;\ \mathbf{c}_b]\right) + \mathbf{b}^{\text{rec}}_{s(b)} \in \mathbb{R}^{n_{\text{out}}}, \tag{14}$$

with $\text{MLP}_{\text{rec}} : \mathbb{R}^{328} \to \mathbb{R}^{128} \to \mathbb{R}^{n_{\text{out}}}$; it replaces the manuscript's perturbed-gene mean in the interaction head by the mark-key mean of Eq. 10.

The per-dataset term is a bias. The design calls for a per-dataset affine at the head, $a_s f(\cdot) + \mathbf{b}_s$. `CellModel043` builds the gene head with `per_dataset_affine=False` and adds $\mathbf{b}_s$ alone, so the scale a_s is fixed at 1 in every run so far. Under HL-Gauss $\mathbf{b}_s$ adds to the 32 logits, a learned prior histogram per dataset. The dataset id therefore enters in two places, both at the heads: $\mathbf{E}^{\text{data}}$ inside $\mathbf{c}_b$ and the bias. It never reaches the trunk or the operator.

4 Objective, step by step ✗

4.1 Targets ✗

Step 1, the transform. Each dataset's released value is first put on a scale where differences are comparable, $\tilde{y}_{b,i} = \text{tr}_s(y_{b,i})$: as released for Kemmeren (already a $\log_2$ ratio to wild type) and Vaishnav (a sort-seq level); the $\log_2$ ratio to the reference for Messner; the difference to the reference for Zelezniak; $\log_2(y+1)$ of TPM for Caudal; $\log_2 y$ of positive values for Muenzner protein and turnover; and $\log_2(10^6 y/\text{UMI total} + 1)$ for the Boocock and N'Guessan counts, where a gene a record omits is a released zero. The primer's table of output scales gives the reasons per dataset.

Step 2, standardization per dataset. On the training rows of dataset s only,

$$z_{b,i} = \frac{\tilde{y}_{b,i} - \eta_{s,i} - \mu_s}{\sigma_s}, \qquad m_{b,i} = \mathbf{1}\left[z_{b,i} \text{ is finite}\right], \tag{15}$$

where $\eta_{s,i}$ is the mean of gene i over the training rows that measure it for the five panel datasets centered per gene (Caudal, Muenzner protein and turnover, Boocock, N'Guessan) and zero for the rest, and μ_s, σ_s are the mean and standard deviation pooled over every centered training value of s. A gene with no training value has no center, so its z is missing rather than zero. The fitted statistics are one pydantic record pinned by sha256 (`0d5df0f0...` in the overnight runs, job 6136 log). Counts never detected in a Boocock or N'Guessan training row are not targets.

4.2 Elementwise losses ✗

Step 3, HL-Gauss bins. The main loss (decision O4 of the primer) bins z into $n_{\text{bin}} = 32$ bins on $[-4, 4]$ of width $\Delta = 0.25$, with interior edges $v_u = -4 + u\Delta$ for $u = 1, \ldots, 31$, the outer edges opened to $v_0 = -\infty$ and $v_{32} = +\infty$, and nominal centers $\bar{v}_u = -4 + (u - \frac{1}{2})\Delta$.

Step 4, the Gaussian histogram target. A measured z becomes the mass a normal $\mathcal{N}(z, \sigma^2_{\text{HL}})$ puts in each bin, with $\sigma_{\text{HL}} = 0.75\Delta = 0.1875$ and Φ the standard normal distribution function,

$$\omega_u(z) = \Phi\Big(\frac{v_u - z}{\sigma_{\text{HL}}}\Big) - \Phi\Big(\frac{v_{u-1} - z}{\sigma_{\text{HL}}}\Big), \qquad u = 1, \ldots, n_{\text{bin}}, \qquad \sum_u \omega_u(z) = 1. \tag{16}$$

Because the edge bins are open, a target beyond ± 4 still has a histogram that sums to one; its mass sits in the edge bin.

Step 5, the cross-entropy. The head's $n_{\text{out}} = 32$ outputs are logits,

$$\hat{\omega}_u = \frac{\exp(\hat{o}_u)}{\sum_{u'} \exp(\hat{o}_{u'})}, \qquad \mathcal{L}^{\text{HL}}(\hat{\mathbf{o}}, z) = -\sum_{u=1}^{n_{\text{bin}}} \omega_u(z) \log \hat{\omega}_u. \tag{17}$$

Step 6, the point prediction. The prediction scored by every correlation is the expectation over the nominal centers,

$$\hat{z} = \sum_{u=1}^{n_{\text{bin}}} \hat{\omega}_u \, \bar{v}_u \in [-3.875,\ 3.875], \tag{18}$$

a convex combination of the centers, so it lies inside the outermost centers and moves toward the middle of the histogram when the predicted mass spreads. The trainer also logs the entropy of $\hat{\omega}$ and the fraction of measured entries whose argmax bin is within one bin of the target's.

Step 7, the pinball arm. The quantile arm emits $n_{\text{out}} = n_q = 19$ values at levels $\gamma_u = u/20$ and averages the check loss,

$$\mathcal{L}^{\text{pin}}(\hat{\mathbf{o}}, z) = \frac{1}{n_q} \sum_{u=1}^{n_q} \max\big(\gamma_u(z - \hat{o}_u),\ (\gamma_u - 1)(z - \hat{o}_u)\big), \qquad \hat{z} = \hat{o}_{10}, \tag{19}$$

the point being the median quantile ($\gamma_{10} = 0.5$). Nothing orders the 19 outputs, so quantile crossing is not prevented.

Step 8, the MSE reference. With $n_{\text{out}} = 1$, $\mathcal{L}^{\text{MSE}}(\hat{o}, z) = (\hat{o} - z)^2$ and $\hat{z} = \hat{o}$. The overnight queue dropped its run for time (Sec. 6).

4.3 From elements to the batch objective ✗

Step 9, the mean within a dataset. Write $\mathcal{L}^{\circ}$ for whichever elementwise loss the arm uses. For dataset s in batch $\mathcal{B}$, the loss is the mean over its measured entries, pooled over its records,

$$\mathcal{L}_s(\mathcal{B}) = \frac{\sum_{b \in \mathcal{B},\, s(b)=s} \ \sum_i m_{b,i} \, \mathcal{L}^{\circ}(\hat{\mathbf{o}}_{b,i}, z_{b,i})}{\sum_{b \in \mathcal{B},\, s(b)=s} \ \sum_i m_{b,i}}, \tag{20}$$

where i runs over the N genes of a gene-level record and over its single entry for a record-level one. Within a dataset a record with more measured genes therefore counts for more.

Step 10, the weighting across datasets. With $\mathcal{S}_{\mathcal{B}}$ the datasets that have a measured entry in the batch,

$$\mathcal{L}_y(\mathcal{B}) = \sum_{s \in \mathcal{S}_{\mathcal{B}}} w_s(\mathcal{B}) \, \mathcal{L}_s(\mathcal{B}), \qquad w_s(\mathcal{B}) = \begin{cases} 1/|\mathcal{S}_{\mathcal{B}}| & \text{equal weighting (main),} \\ n_s(\mathcal{B}) \big/ \sum_{s' \in \mathcal{S}_{\mathcal{B}}} n_{s'}(\mathcal{B}) & \text{record weighting (arm).} \end{cases} \tag{21}$$

Against the manuscript's $\mathcal{L}_y = \sum_t w_t \sum_b m_t^{(b)} \ell_t$, two things differ: each dataset's term is normalized by its own measured count, and the weights are not constants but functions of what the batch happens to hold. 043 adds no graph term, so $\mathcal{L} = \mathcal{L}_y$.

4.4 What a batch holds ✗

Step 11, the records sampler. The default sampler shuffles the $n = 50{,}767$ training records each epoch and cuts the order into batches of $B = 8$, so a batch is a draw without replacement and its composition $(n_s(\mathcal{B}))_s$ is multivariate hypergeometric (the last batch of an epoch holds 7 records, $50{,}767 = 6{,}345 \times 8 + 7$; it is ignored below). Vaishnav holds $n_{\text{vai}} = 42{,}694$ of them, 84.1%, so a batch holds $8 \times (1 - n_{\text{vai}}/n) = 1.27$ gene-level records on average.

Step 12, the expected weight as a gradient share. The batch gradient is $\nabla \mathcal{L}_y = \sum_s w_s(\mathcal{B}) \nabla \mathcal{L}_s(\mathcal{B})$, so the expected weight

$$\bar{w}_s = \mathbb{E}_{\mathcal{B}}\big[w_s(\mathcal{B})\big] = \sum_{(n_1, \ldots, n_9)} \frac{\prod_{s'} \binom{n_{s'}}{n_{s'}(\mathcal{B})}}{\binom{n}{B}} \, w_s(\mathcal{B}), \qquad \bar{w}_s^{\text{records}} = \frac{n_s}{n}, \tag{22}$$

is the expected share of the objective's gradient that dataset s carries, if the per-dataset gradients $\nabla \mathcal{L}_s$ have comparable norms. The sum runs over every composition of a batch of 8 into the nine datasets, and the record-weighting value follows from linearity. Table 2 gives both exactly. Under equal weighting Vaishnav carries $\bar{w}_{\text{vai}} = 0.573$: it is present in nearly every batch, and the batches that also hold a gene-level record split the weight only two or three ways. Kemmeren

is present in a fraction 0.174 of batches and carries $\bar{w} = 0.066$. Hypothesis (untested): the per-dataset gradient norms are not comparable, since the 32-bin cross-entropy of one Vaishnav scalar and the mean over about 6,000 genes of a Kemmeren profile differ in variance; the norms were not logged.

Step 13, the balanced sampler. The alternative draws, for each of the 8 slots of a batch, a dataset uniformly among the nine and then a record uniformly within it, with replacement, for the same 6,346 batches per epoch. The composition is then multinomial with equal cell probabilities, and both weightings give $\bar{w}_s = 0.111$ for every dataset by symmetry. Each dataset is drawn about $6{,}346 \times 8/9 \approx 5{,}641$ times per epoch, so the 33 turnover training records are each revisited about 171 times per epoch and Vaishnav's 42,694 are subsampled to about 13%. The sampler acts on the batch and the weighting on the loss; the two compose, and a balanced run is labeled `<arm>_balanced_<loss>_<weighting>`.

Table 2: Training records and the expected weight of each dataset in the batch objective, `joint_9`, batches of 8. Records from the prepared store (job 6134). Validation records are those the trainer scores per evaluation (at most 2,000 per dataset). Expected weights are exact (Eq. 22): the records sampler is a draw without replacement (multivariate hypergeometric over all 12,870 compositions of 8 into nine datasets), the balanced sampler a multinomial with probability 1/9 per dataset. Probabilities and weights are per batch.

Dataset	Level	Training records, n_s	Validation records scored, n	Share of training records, %	Probability present in a batch, records sampler	Expected weight $\bar{w}_s$, records sampler, equal weighting	Expected weight $\bar{w}_s$, records sampler, record weighting	Expected weight $\bar{w}_s$, balanced sampler, either weighting
Kemmeren 2014	gene	1,202	140	2.4	0.174	0.066	0.024	0.111
Messner 2023	gene	3,712	455	7.3	0.455	0.188	0.073	0.111
Zelezniak 2018	gene	79	9	0.2	0.012	0.004	0.002	0.111
Caudal 2024	gene	566	189	1.1	0.086	0.032	0.011	0.111
Muenzner 2024 protein	gene	476	159	0.9	0.073	0.027	0.009	0.111
Muenzner 2024 turnover	gene	33	11	0.1	0.005	0.002	0.001	0.111
Boocock 2025 pooled	gene	235	79	0.5	0.036	0.013	0.005	0.111
N'Guessan 2025	gene	1,770	591	3.5	0.247	0.095	0.035	0.111
Vaishnav 2022	record	42,694	2,000 of 14,232	84.1	1.000	0.573	0.841	0.111
All nine		50,767	3,633	100.0		1.000	1.000	1.000

4.5 Optimization and the validation loss ✗

Step 14, the optimizer. AdamW at learning rate 3×10^{-4} with weight decay 10^{-8}, linear warmup over 500 steps then constant, gradient norm clipped at 10, dropout 0.1, bf16 autocast on the card. A non-finite gradient norm stops the run.

Step 15, the validation loss. The validation pass computes the same objective with the same weighting, but accumulates the per-dataset sums of Eq. 20 over the whole pass before combining them, so under equal weighting

$$\mathcal{L}^{\text{val}} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \frac{\sum_{b \in \text{val}_s} \sum_i m_{b,i}\, \mathcal{L}^{\circ}(\hat{\mathbf{o}}_{b,i}, z_{b,i})}{\sum_{b \in \text{val}_s} \sum_i m_{b,i}}, \tag{23}$$

logged as `val/loss` with each dataset's term as `val/<dataset>/loss`. The training curve `train/loss` is the mean of $\mathcal{L}_y(\mathcal{B})$ over the epoch's batches, whose weights follow Table 2; the two are the same loss but not the same weighting of datasets, so their gap mixes generalization with the batch composition. The validation loss was added on 2026-10-23 (commit `e6ad22d0c`); none of the overnight runs of 2026-10-22 has it, and the prepare-experiment gate (`.claude/skills/prepare-experiment/SKILL.md`) now refuses a chain whose smoke run lacks `val/loss` and the per-dataset pair.

5 Evaluation and the decision rule ✗

5.1 The split ✗

One fold file, `conf/splits/global_split_v1.json` (seed 0, job 6046), gives every record a role. It holds one fold id per key in four key spaces: the 4,908 deletion genes in 10 folds, stratified over 38 strata of dataset membership (which

deletion datasets measure the gene); the 979 isolates by strain name in 5 folds, stratified over 4 membership strata; the segregant records in 5 folds drawn within each cross and panel; and the 71,158 distinct promoter 80-mers in 5 folds. Fold 0 is test, fold 1 validation, the rest training. A deletion record trains only when none of its genes is in fold 0 or 1, is test when any is in fold 0, and is validation otherwise; the 83 pinned genes (the 82 Sameith 2015 singles and the HO control of Jackson 2020) take folds 2 to 9 only. Three datasets have a fixed role: Sameith 2015 Dm is split B, read through Kemmeren's dataset token and standardization because it comes from the same arrays; Ohya 2005 is the morphology probe; Sameith 2015 Sm is excluded. The prepare stage drops fold 0, so its labels are never loaded during development, and the doubles, logged as `val/doubles/...`, are a test set that must not steer a choice.

5.2 Statistics ✗

Per-gene Pearson. For a gene-level dataset and its held-out records $\mathcal{V}$, the primary statistic is the median over genes of the Pearson correlation across records,

$$r_i = \mathrm{corr}_{b\in\mathcal{V}}\left(\hat{z}_{b,i},\ z_{b,i}\right), \qquad \mathrm{stat} = \underset{i}{\mathrm{median}}\ r_i, \tag{24}$$

over genes measured in at least 3 held-out records with nonzero variance on both sides. The standardization of Eq. 15 is affine per gene, so r_i is the same on z and on the transformed scale $\tilde{y}$.

Centered per-strain Pearson. With $\bar{z}_i$ the training-mean profile of the dataset,

$$r_b = \mathrm{corr}_i\left(\hat{z}_{b,i} - \bar{z}_i,\ z_{b,i} - \bar{z}_i\right), \qquad \mathrm{stat} = \underset{b\in\mathcal{V}}{\mathrm{median}}\ r_b. \tag{25}$$

Centering removes the shared deletion response that a model emitting the training mean would score on: two different Kemmeren mutants already agree at a raw 0.284 (job 5893, primer Sec. 3.1).

Vaishnav. One value per promoter, so the statistic is the Pearson correlation of $\hat{z}_b$ and z_b over held-out promoters.

The Sameith doubles. The 72 doubles are scored in released units, through the inverse of Kemmeren's standardization, centered per double on Kemmeren's training-mean profile $\bar{y}_i$, over all 6,127 genes,

$$r_{xx'} = \mathrm{corr}_i\left(\hat{y}_{xx',i} - \bar{y}_i,\ y_{xx',i} - \bar{y}_i\right), \qquad \mathrm{stat} = \underset{xx'}{\mathrm{median}}\ r_{xx'}, \tag{26}$$

for the double deletion of genes x and x'. Two baselines need no model: the additive null of the measured singles, $\hat{y}_{xx'} = y_x + y_{x'}$ (the paper's own expected value for a log ratio), and the fitted rung

$$\hat{y}_{xx'} = \lambda_1 y_{\mathrm{strong}} + \lambda_2 y_{\mathrm{weak}} + \lambda_0, \tag{27}$$

fitted leaving each double out, with y_{strong} the single of larger effect; its mean coefficients are $\lambda_1 = 0.638$, $\lambda_2 = 0.312$ and $\lambda_0 = 0.005$ (`results/linear_evals_043.json`, job 6060). The trainer also logs the model's own additive prediction, the sum of its predictions for the two singles.

5.3 The registered window ✗

The read-out is declared in the config before launch. For a run of n_{ep} epochs,

$$\mathcal{W} = \left\{\lfloor 0.8\, n_{\mathrm{ep}}\rfloor + 1, \ldots, n_{\mathrm{ep}}\right\}, \qquad \overline{\mathrm{stat}} = \frac{1}{|\mathcal{W}_{\mathrm{eval}}|}\sum_{e\in\mathcal{W}_{\mathrm{eval}}} \mathrm{stat}_e, \tag{28}$$

where $\mathcal{W}_{\mathrm{eval}} \subseteq \mathcal{W}$ are the epochs at which validation ran. The 50-epoch joint runs give $\mathcal{W} = \{41, \ldots, 50\}$ with 10 evaluations; the step-matched Kemmeren run (2,102 epochs, validation every 42) gives epochs 1,682 to 2,102 with 11. Validation is computed and logged every evaluation epoch and never selects an epoch: a maximum over noisy epochs is an upward-biased order statistic, and the window mean is not. One implementation detail differs from the intent: checkpoints are written at the window's first and last epoch only, not at every epoch inside it (`train_043.py`, `epoch in window` on a tuple).

5.4 Transfer at matched optimizer steps ✗

Transfer to a dataset is the joint run's window mean minus the window mean of a run trained on that dataset alone, with the same seed and the same number of optimizer steps,

$$\Delta_s^{\mathrm{tr}} = \overline{\mathrm{stat}}_s^{\,\mathrm{joint}} - \overline{\mathrm{stat}}_s^{\,\mathrm{alone}} \quad \text{at equal } n_{\mathrm{step}}. \tag{29}$$

For Kemmeren the arithmetic is

$$\text{joint:}\quad \lceil 50{,}767/8 \rceil = 6{,}346 \text{ steps per epoch}, \qquad 6{,}346 \times 50 = 317{,}300 = n_{\text{step}}, \tag{30}$$

$$\text{alone:}\quad \lceil 1{,}202/8 \rceil = 151 \text{ steps per epoch}, \qquad \lceil 317{,}300/151 \rceil = \lceil 2{,}101.3 \rceil = 2{,}102 \text{ epochs}, \tag{31}$$

the last epoch cut to 49 steps so the total is exactly $2{,}101 \times 151 + 49 = 317{,}300$ (`budget.max_steps`, commit `694b80513`); a time budget that would cut the alone run short raises instead of training an unmatched control. Matching on epochs would give the alone run 50 epochs of 151 steps, 7,550 optimizer steps, 2.4% of the joint run's.

5.5 The decision rule ✗

An arm is called better than its named control on a split when all of the following hold, with the control named before the first read: the mean paired gain in the primary statistic's window mean is at least 0.02; the gain is positive in at least 5 of 6 paired initialization seeds; and the 95% interval of the gain, resampled over held-out strains (over genes for a per-gene statistic), excludes zero. The test fold is read once, at the end of a campaign. The primer's protocol (Sec. 5.1) calls for the rule to be committed as `conf/decision_rule.yaml` before the first GPU run; no such file is in `conf/` at commit `bed26f949`, so the rule stands in the primer and here only. Hypothesis (untested): 6 seeds suffice to separate a 0.02 gain on Kemmeren from seed noise; experiment 031 found the seed standard deviation equal to the pinned fold's sampling standard deviation (0.041), so the seed spread of 043 has to be measured before the rule's power is known.

6 What has been measured ✗

The overnight round of 2026-10-22 ran five jobs on two GilaHyper cards, all seed 0, all on the prepared store of job 6134 (Table 3). Four completed. Table 4 sets their registered-window means against the linear ladder of `results/linear_evals_043.json`, read on fold 1, the fold the runs validate on, and on fold 0, where the ladder was first reported. The ladder fits on folds other than its test fold, so on fold 1 it trains on fold 0 as well; the held-out units are the same as the runs'. Every number here is one seed, below the decision rule's six, and none of these runs logged a validation loss: the only loss curve is `train/loss`.

Table 3: The overnight runs of 2026-10-22, seed 0. Each run id links to its W&B run (project `torchcell_043-expression-proteome-morphology`). Epochs are those in the W&B history when the window means were computed; the registered window is the last 20% of each run's epochs.

Slurm job	W&B run	Arm (W&B group)	Epochs reached of planned, n	Registered window, epochs	Validation evaluations in the window, n
6136	g5wda4mp	`joint_9_hlgauss_equal`	50 of 50	41 to 50	10
6137	cqdm487x	`joint_9_balanced_hlgauss_equal`	50 of 50	41 to 50	10
6150	5f3ijxa1	`joint_9_pinball_equal`	50 of 50	41 to 50	10
6152	kmlepsr2	`alone_kemmeren_hlgauss_equal`	2,102 of 2,102	1,682 to 2,102	11
6151	2391fl7t	`joint_9_hlgauss_records`	26 of 39	32 to 39	window not reached

Readings. On Kemmeren the joint run with the records sampler reads 0.078 against 0.110 alone at matched steps, a transfer of −0.032; the balanced sampler reads 0.119, a transfer of +0.009; the pinball arm −0.020. All four Kemmeren values sit below the fold-1 ridge on the deleted gene's frozen composite token. The doubles read 0.208 to 0.230 jointly and 0.335 alone, all below the additive null of the measured singles (0.440). On the isolates every joint run is below the GBLUP bar, the balanced sampler closest (Muenzner 0.278 against 0.471, Caudal 0.175 against 0.372). Messner reads 0.059 to 0.065 against 0.082. Vaishnav is the one readout at or above its bar: 0.921 with the records sampler against a k-mer ridge of 0.885, on a 2,000-promoter subset. Hypothesis (untested): the balanced sampler's higher Kemmeren, Caudal and Muenzner means and lower Vaishnav mean follow from the shift in expected weight of Table 2 (Kemmeren from 0.066 to 0.111, Vaishnav from 0.573 to 0.111); one seed cannot separate that from seed noise.

The canceled run. Job 6151 (`joint_9_hlgauss_records`, run `2391fl7t`) is the record-weighting arm. Its trainer fixed 39 epochs from the time left before the end of the owner's card grant (07:29), so its registered window was epochs 32 to 39. It was canceled by hand at 06:33:55 on 2026-10-23, after 1 h 56 min, while in epoch 32: its log shows 31 completed epochs, and the W&B history read for Table 3 held 26. The window was never reached, so no number is reported for it. The MSE reference run was dropped from the queue before launch because the second card's chain could not reach it before the end of the granted window.

Run links. The four complete runs and the canceled one, in the order of Table 3:

Table 4: Registered-window means of the overnight runs against the linear bars, one seed. Each run column is the mean of the row's statistic over the evaluations inside the run's registered window, on the validation fold (fold 1), seed 0, from the unsampled W&B history. Bold marks the highest of the four runs in each row. The ladder columns are the fold-1 and fold-0 values of the named linear model on the same held-out units (`linear_evals_043.py`, jobs 6059 to 6067); the doubles are split B, one set, so the two doubles bars are the additive null of the measured singles and the fitted rung of Eq. 27. The runs score Vaishnav on a fixed subset of 2,000 validation promoters, the ladder on all of them. The last row is Eq. 29; a single seed cannot meet the rule it names.

Readout and statistic	Window mean, Kemmeren alone (6152)	Window mean, joint, records sampler (6136)	Window mean, joint, balanced sampler (6137)	Window mean, joint, pinball loss (6150)	Linear bar, same statistic, fold 1	Linear bar, same statistic, fold 0	Linear model of the bar
Kemmeren 2014, median per-gene Pearson r over 140 held-out deletion strains	0.110	0.078	**0.119**	0.090	0.131	0.125	ridge, composite token of the deleted gene
Sameith 2015 doubles, median over 72 doubles of the centered per-double Pearson r (6,127 genes)	**0.335**	0.208	0.218	0.229	0.440 / 0.494	same set (split B)	additive measured / fitted, leave one out
Messner 2023, median per-gene Pearson r over 441 held-out deletion genes	–	0.059	0.064	**0.065**	0.082	0.083	ridge, composite token of the deleted gene
Caudal 2024, median per-gene Pearson r over 189 held-out isolates	–	0.155	**0.175**	0.160	0.372	0.369	GBLUP, kinship from gene-level calls
Muenzner 2024 protein, median per-gene Pearson r over 159 held-out isolates	–	0.202	**0.278**	0.210	0.471	0.423	GBLUP, kinship from gene-level calls
Vaishnav 2022, Pearson r over held-out promoters (runs: 2,000 of 14,232; ladder: all)	–	**0.921**	0.866	0.914	0.885	0.884	ridge, k-mer counts (k 1 to 6)
Transfer on Kemmeren 2014, joint minus alone at 317,300 optimizer steps	0 (reference)	−0.032	+0.009	−0.020	–	–	rule: mean gain ≥ 0.02, 5 of 6 seeds up

https://wandb.ai/zhao-group/torchcell_043-expression-proteome-morphology/runs/g5wda4mp
https://wandb.ai/zhao-group/torchcell_043-expression-proteome-morphology/runs/cqdm487x
https://wandb.ai/zhao-group/torchcell_043-expression-proteome-morphology/runs/5f3ijxa1
https://wandb.ai/zhao-group/torchcell_043-expression-proteome-morphology/runs/kmlepsr2
https://wandb.ai/zhao-group/torchcell_043-expression-proteome-morphology/runs/2391fl7t

7 Open design questions ✗

Each question names the choice, the fact that bears on it, and the measurement that would settle it. Arms are compared under the rule of Sec. 5.5, on the window mean of Eq. 28, paired by seed.

1. **Heterozygous IUPAC codes: strict gap or the encoder's unknown token.** An isolate input with a character outside its encoder's alphabet is the typed gap `ambiguous_sequence` today. The rule gaps 2.4% of isolate-gene rows for FUDT upstream, 1.2% for FUDT downstream, 20.8% for CaLM and 16.1% for ProtT5 (job 6126); FUDT and ProtT5 would accept the same inputs by reading the odd character as their unknown token. *Test:* embed the gapped rows through the unknown token, rebuild the isolate representation, and compare the strict-gap and unknown-token arms on Caudal and Muenzner per-gene Pearson, six seeds each; report also how far each gapped protein's unknown-token vector lies from its gene's S288C vector.

2. **Soft-masked bases.** The lowercase a, c, g, t, n of the isolate assemblies are soft-masked bases, not ambiguities, but fall under the same rule (about 3% of the distinct untokenizable windows). Uppercasing at slice time would change the stored window string. *Test:* count the windows the uppercase slice recovers, re-embed them (an align and merge rerun of a few minutes), and compare the recovered and strict-gap arms on Caudal and Muenzner per-gene Pearson.

3. **Route B, and the consistency test between routes.** Route B reads an isolate on S288C with its variant and loss-of-function genes as marks (a median of 5,038 marks per Caudal record, against at most one engineered mark under route A; job 6120). It is not implemented in the trainer. *Test:* the two routes on the same isolates and seeds, compared on the isolate statistics; and a consistency check that needs no labels: on held-out isolates, the correlation between the two routes' predicted profiles, against the correlation of either with the measurement.

4. **Attended-token head or head-only tokens.** $\mathbf{c}_b$ reaches only the heads (Eqs. 13, 14). The attended form (form c of the primer) also offers the tokens as keys to the operator, so the experiment type could change which genes move. *Test:* the attended arm against head-only on Kemmeren, the doubles and the isolates, six seeds; not implemented.

5. **Flat-type and six-heads controls.** The type path of Eq. 11 assumes that sharing a group row helps. *Test:* `exp_type_mode: flat` (one row per leaf) and six heads, one per experiment type with no shared group, each against the DAG arm on every dataset; the flat control needs only the config switch, the six-heads control is not implemented.

6. **Dataset weighting extremes.** Table 2 spans an expected Vaishnav weight of 0.111 (balanced) to 0.841 (records sampler, record weighting); the record-weighting run (6151) was canceled before its window. *Test:* the full two-by-two of sampler (records, balanced) and weighting (equal, records), six seeds, read on Kemmeren, the doubles, the isolates and Vaishnav, with `val/loss` logged (Eq. 23).

7. **A clade-disjoint isolate split.** The random isolate folds leave relatives of a held-out isolate in training, so the GBLUP bars are the easy-split bars. *Test:* a second fold file disjoint by Peter 2018 clade, beside `global_split_v1`; GBLUP and the model on both, with the drop from random to clade split as the quantity of interest.

8. **The Addgene 127546 window gate.** The Vaishnav reporter's upstream key is a 1,003 bp window of plasmid 127546. Every GenBank and SnapGene download of the plasmid returned HTTP 404, so the window comes from a reconstruction of 8,218 bp that matches the stated length and five ORFs without internal stops (`results/addgene_127546_fetch.json`); all 71,158 windows were embedded on it (job 6115). *Test:* deposit the GenBank file by the recorded manual recipe, compare its sha256-pinned window with the reconstruction base by base, and re-embed only on a difference; until then, decide whether the Vaishnav readings of Table 4 may be quoted.

9. **Retiring Sameith 2015 Sm.** Its 82 profiles are Kemmeren's own arrays (82 of 82 identical at $r = 1.000$ over 6,169 genes), so the split excludes it. The question is whether it leaves the served store or stays as a typed duplicate. *Test:* repeat the identity check on the served graph's records rather than the raw store, and record the result with the retirement decision.

10. **The 188 dropped records.** The deletion blocks of the 043 query use an `ALL` clause over the genome gene set, which drops 188 released records whose gene name is outside the 6,607-gene index: Kemmeren 2, Hu 1, Nadal-Ribelles 70, Messner 75, Ohya 40. The database constitution keeps every released record and lets the genome gate nothing. *Test:* re-query with the isolates' `SIZE = 0 OR ANY` shape (about 6 h), confirm the store gains exactly 188 records,

and report their names under the name-currency rule; whether they train then depends on whether their genes have a token.

11. **The mark cap.** The overnight runs kept at most 1,024 marks per record, a smoke-time cap; N'Guessan carries a median of 2,910 and Boocock de novo cells 3,351. *Test:* an uncapped run's memory and step time on one card, then the capped and uncapped arms on the segregant statistics.